Zheyuan Dai
Madison, WI

dzy040811@gmail.com
+1 (909) 721-4701
linkedin.com/in/zheyuan-dai-b7b108382

Research Interests

Statistical machine learning, online learning, differential privacy, graph algorithms, and generative modeling.

Education

University of Wisconsin–Madison Madison, WI
M.S. in Mathematics, expected May 2027 *Sep. 2026 – May 2027*
– Visiting International Student in Mathematics, Sep. 2025 – May 2026.

Shanghai Jiao Tong University Shanghai, China
B.S. in Mathematics and Applied Mathematics. *Sep. 2022 – Jun. 2026*

Honors

Outstanding Undergraduate Thesis Award, School of Mathematical Sciences, Shanghai Jiao Tong University, 2026.

Second Prize and Best Creativity Award, VEX Mechanical Competition, Shanghai Jiao Tong University, Nov. 2023.

Publications

Zheyuan Dai, Xuliang Zhu, Yizhang He, Kai Wang, Fan Zhang, and Xuemin Lin. "Robust Coreness Estimation via H-index under Edge-LDP." Accepted at **ACM SIGMOD 2027**.

Working Papers

Zheyuan Dai, Xuliang Zhu, Ruijia Wu, Yizhang He, Gengda Zhao, and Kai Wang. "Private Clicks, Public Actions: Logistic Bandits under Local Differential Privacy." Manuscript in preparation.

Zheyuan Dai, Xuliang Zhu, and Kai Wang. "Batch Caching with Overlapping Costs: Complexity and Approximation." Manuscript in preparation.

Research Experience

Sharp Policy Evaluation under State Expansion Sep. 2026 – Present
Advisor: Prof. Yongyi Guo, University of Wisconsin–Madison
– Developing sharp identification bounds for offline policy evaluation in finite-horizon Markov decision processes when evaluation policies use newly available state variables absent from historical decision rules.
– Characterizing compatible full-data laws through measurable conditional couplings and deriving attainable policy-value bounds via backward dynamic programming and rearrangement inequalities.

Private Graph Algorithms and Online Learning 2025 – Present
Advisors: Prof. Xuliang Zhu and Prof. Kai Wang
– Developed and analyzed iterative H-index estimators for coreness estimation under edge-level local differential privacy, with approximate k-core guarantees and evaluation on large real-world graphs.
– Developed frequentist and Bayesian bandit algorithms with regret guarantees for private online recommendation under reward-only local differential privacy.

Leadership and Activities

Shanghai Jiao Tong University Mountaineering Association Shanghai, China
President *Jun. 2023 – Jun. 2025*
– Led the university mountaineering team and organized technical training, expedition planning, and outdoor activities.

Skills

Programming: Python, C++, MATLAB, R, LaTeX **Libraries:** PyTorch, pandas, NumPy, NetworkX

Languages: Chinese (native), English (TOEFL 102; CET-6)